

Diogo Vieira da Silva Pellegrina, PhD

+1 647 512 8681 diogo.pellegrina@gmail.com linkedin.com/in/diogo-pellegrina-187634178/

Postdoc in bioinformatics;
Modeling, statistical analysis, machine learning, and visualization of
all types of biological data.

Skills and Keywords

Programming Languages **Expert in Python, R, Bash**; Good in PostgreSQL, Java, C; Had experiences with C++, MATLAB, AWK.

CS and Statistics **Data Visualization, High Performance Computing (HPC)**, Parallel processing, Machine Learning, Clustering, Linear and Non-Linear Regression Modeling, Data Science, Linear Algebra, Git

Biology **Genetics, Immunology, Next Generation Sequencing (NGS)**, Transcriptome (RNAseq), Whole Genome Sequencing (NGS), Exome, Copy Number Alterations, ATAC-seq, single cell, Network Interactions.

Professional experience

Instructor for the Canadian Bioinformatics Workshops

- Lecturing a class in RNA-seq Analysis, Pathway Analysis and Network Analysis.

Postdoc at the Vaccine and Infectious Disease Organization:

- **Development of an algorithm** to refine pathway analysis.
- **Standardization of bioinformatics pipelines** for NGS workflows such as Proteomics, Metabolomics, Chip-Seq, and single-cell RNAseq analysis.
- Preparation of a grant proposal for designing vaccine adjuvants.

Postdoc at the Ontario Institute for Cancer Research:

- Proposed, **acquired external funding**, and developed analysis of COVID phosphorylation networks.
- Statistical analysis of Mutational Signatures and Copy Number Alterations

Bioinformatics Masters and PhD at the Institute of Mathematics and Statistics at the University of Sao Paulo:

- Classification of regulatory elements using Weighted **Co-expression Networks** Analysis (WGCNA).
- **Integration** of mutation and gene expression on cancer datasets.

150+ citations, 10 publications as first Author

Complete list of publications and statistics on Google Scholar:

scholar.google.com/citations?user=b3uvtk4AAAAJ

Detailed Information:

2024 – Current	Post-doc research fellow at the Vaccine and Infectious Disease Organization
2020 – 2024	Post-doc research fellow at the Ontario Institute for Cancer Research (OICR). Advised by: Jüri Reimand
2016 – 2020	PhD in Bioinformatics at the University of Sao Paulo; <i>"Integration of Genomic and Transcriptomic Analyses in Pancreatic Adenocarcinoma"</i> . Advised by: Eduardo M. Reis
2013 – 2016	Master's in Bioinformatics at the University of Sao Paulo; <i>"Septic shock in advanced age: Transcriptome analysis reveals altered molecular signatures in neutrophil granulocytes"</i> . Advised by: Eduardo M. Reis
2014 – 2019	Teaching assistant in Introduction to Programming, and in Algorithm Development courses for 1st year students of Molecular Sciences and for 1st year students of Applied Mathematics at the University of Sao Paulo (USP)
2010 – 2013	Undergraduate Research at the Physics Institute; <i>"Phase transition in lipidic bilayers: an spin label structural study"</i> . Advised by: Maria Teresa Lamy

Complementary Courses

2019	São Paulo School of Computational Science on Learning from Data, University of São Paulo (USP). A course focused on HPC, Machine Learning, Pattern Recognition and Database Management.
2017	Internship at Ohio State University, United States, developing tumor sequencing pipelines under the supervision of Profs. David Symer and Keiko Akagi.
2016	1st Summer School of Bioinformatics at Universidade Federal de São Paulo (UNIFESP). A collaboration of UNIFESP with King's College London, the classes focused on GWAS, Pathway Analysis, Gene expression, WGCNA, and Methylation data.
2010	Mathematical Methods in Population Biology III, State University of São Paulo (UNESP). During this course I developed software based on stochastic models to simulate the growth and ecological stability of Dengue virus based on qualitative and quantitative traits. Of particular relevance since climate change is expanding Dengue's endemic range.

Conferences awards

2025	Selected Speaker at the 2025 Bioinformatics Network (Bionet) conference in Calgary, Alberta
2023	Selected Speaker at the 2023 International Conference in Systems Biology in Hartford, Connecticut
2018	Best poster at the 2018 Bioinformatics Workshop and summer course at USP
2013	At the 21 ^o SIICUSP. Final version of “ <i>Phase transition in lipidic bilayers: an spin label structural study</i> ” was awarded as one of the 3% best undergraduate works of the year in the state of São Paulo.
2011	At the 19 ^o SIICUSP. Preliminar version of “ <i>Phase transition in lipidic bilayers: an spin label structural study</i> ”, this work was evaluated as one of the 5 best in Physics, and therefore selected to a special oral presentation.

Fellowship Awards

2022	My post-doc project “ <i>Integrative analysis of population polymorphism interference in host phospho-signaling response networks to SARS-CoV-2 infection</i> ” was awarded by the 2022 CANSSI Ontario STAGE HostSeq Project Fellowship competition.
2016	After the 2-year FAPESP grant, the PhD was funded by the 5-year CNPq national graduate scholarship funding.
2014	Masters research project and preliminary results were awarded the FAPESP 2-year graduate scholarship funding, Brazil’s most prestigious and highest stipend scholarship.
2012	Due to the award at the 19 ^o SIICUSP Symposium I received an CNPq national undergraduate scholarship funding

Languages

Portuguese (*native*), English (*CELP* 45/48), and French (*DELF A1*).

Publications as first author

- Pellegrina, D., Helmy, M., **AI for Scientific Integrity: Detecting Ethical Breaches, Errors, and Misconduct in Manuscripts**. Front. Artif. Intell. (2025) doi: 10.3389/frai.2025.1644098
- Pellegrina, D., Wilson, H. L., Mutwiri, G. K., & Helmy, M. (2025). **Transcriptional Systems Vaccinology Approaches for Vaccine Adjuvant Profiling**. Vaccines, 13(1), 33.

- Pellegrina, D.*, Prayitno, K.*, A Azhie, E Pasini, C Baciú, S Fischer, J Reimand & M Bhat (2023). **Transcriptomic changes in liver transplant recipients with non-alcoholic steatohepatitis indicate dysregulation of wound healing**. *Frontiers in Endocrinology*, doi:10.3389/fendo.2023.1111614
- Pellegrina, D.*, Bahcheli, A. T.*, Krassowski, M., & Reimand, J. (2022). **Human phospho-signaling networks of SARS-CoV-2 infection are rewired by population genetic variants**. *Molecular Systems Biology*, 18(5), e10823.
- Pellegrina, D. V. D. S., Severino, P., Barbeiro, H. V., De Souza, H. P., Machado, M. C. C., Pinheiro-da-Silva, F., & Reis, E. M. (2017). **Insights into the function of long noncoding RNAs in sepsis revealed by gene co-expression network analysis**. *Non-coding RNA*, 3(1), 5.
- Diogo Vieira da Silva Pellegrina, Patricia Severino, Marcel Cerqueira Machado, Fabiano Pinheiro da Silva, Eduardo Moraes Reis, **Microarray gene expression analysis of neutrophils from elderly septic patients**, *Genomics Data*, Volume 6, December 2015, Pages 51-53, ISSN 2213-5960, <http://dx.doi.org/10.1016/j.gdata.2015.08.013>.
- Vieira da Silva Pellegrina D, Severino P, Vieira Barbeiro H, Maziero Andreghetto F, Tadeu Velasco I, Possolo de Souza H, et al. (2015) **Septic Shock in Advanced Age: Transcriptome Analysis Reveals Altered Molecular Signatures in Neutrophil Granulocytes**. *PLoS ONE* 10(6): e0128341. doi:10.1371/journal.pone.0128341
- DIOGO,VSP ; DUARTE, E. L. ; M. T. Lamy . **Transição de fase de bicamadas lipídicas aniônicas: um estudo estrutural com marcadores de spin..** In: SIICUSP, 2012, São Carlos. Iniciação Científica e Tecnológica: o jovem pesquisador em ação III. São Carlos: CETEPE, 2012. p. 125-138.

Publications as Intermediate author

- Gheybi, Kazzem, Pamela XY Soh, Jue Jiang, Tumisang MN Mbeki, Melanie Louw, Daniel Burns, Piyushkumar Mundra ... & Vanessa M Hayes. **"Pathogenic variants reveal candidate genes for prostate cancer germline testing for men of African ancestry."** *Nature Communications* 16, no. 1 (2025): 8799.
- Bayati, M., Klein, Z.P., Bahcheli, A.T., Slobodyanyuk, M., To, J., Cheng, K.C., Mishra, J., Pellegrina, D., Guevara-Hoyer, K., McIntosh, C. and Bhat, M., 2025. **Cancer genomic alterations and microenvironmental features encode synergistic interactions with disease outcomes**. *Molecular Cancer Research*.
- Adler, N., Bahcheli, A. T., Cheng, K. C., Al-Zahrani, K. N., Slobodyanyuk, M., Pellegrina, D., ... & Reimand, J. (2023). **Mutational processes of tobacco smoking and APOBEC activity generate protein-truncating mutations in cancer genomes**. *Science Advances*, 9(44), eadh3083.

- Pasini E, Baciú C, Angeli M, Arendt B, Pellegrina D, Reimand J, Patel K, Tomlinson G, Mazhab-Jafari MT, Kotra LP, Fischer S. **Acyl-CoA Thioesterase 1 Contributes to Transition of Steatosis to Metabolic-Associated Steatohepatitis.** International Journal of Hepatology. 2024;2024(1):5560676.
- Pillat MM, Ayupe AC, Juvenal G, Meinerz C, Glaser T, da Silva Pellegrina DV, Paiva DS, Mello CF, Longo BM, Reis EM, Ulrich H. **Differentiated embryonic neurospheres from familial Alzheimer's Disease Model Show Innate Immune and glial cell responses.** Stem Cell Reviews and Reports. 2023 Aug;19(6):1800-11.
- Pramio, D. T., Vieceli, F. M., Varella-Branco, E., Goes, C. P., Kobayashi, G. S., da Silva Pellegrina, D. V., ... & Schechtman, D. (2023). **DNA methylation of the promoter region at the CREB1 binding site is a mechanism for the epigenetic regulation of brain-specific PKM ζ .** Biochimica et Biophysica Acta (BBA)-Gene Regulatory Mechanisms, 1866(1), 194909.
- Menezes, P. R., Trufen, C. E. M., Lichtenstein, F., da Silva Pellegrina, D. V., Reis, E. M., & Onuki, J. (2023). **Transcriptome profile analysis reveals putative molecular mechanisms of 5-aminolevulinic acid toxicity.** Archives of Biochemistry and Biophysics, 109540.
- Adler N, Bahcheli AT, Cheng KC, Al-Zahrani KN, Slobodyanyuk M, Pellegrina D, Schramek D, Reimand J. **Mutational processes of tobacco smoking and APOBEC activity generate protein-truncating mutations in cancer genomes.** Science Advances. 2023 Nov 3;9(44):eadh3083.
- da Paixão, V. F., Sosa, O. J., da Silva Pellegrina, D. V., Dazzani, B., Corrêa, T. B., Risério Bertoldi, E., ... & Moraes Reis, E. (2022). **Annotation and functional characterization of long noncoding RNAs deregulated in pancreatic adenocarcinoma.** Cellular Oncology, 1-26.
- Sanches, L., Pato, G. T. G., Pellegrina, D. V. da S., Ferreira, J. E., & Reis, E. M. (2022). **LACEN: an R package for lncRNA functional annotation using co-expression networks.** In Abstract Book. São Paulo: Sociedade Brasileira de Bioquímica e Biologia Molecular.
- Krassowski, M., Pellegrina, D., Mee, M. W., Fradet-Turcotte, A., Bhat, M., & Reimand, J. (2021). **ActiveDriverDB: interpreting genetic variation in human and cancer genomes using post-translational modification sites and signaling networks** (2021 Update). Frontiers in Cell and Developmental Biology, 9, 626821.

Masters and Doctoral thesis

- Pellegrina, D. V. D. S. (2020). Análise integrada de alterações somáticas e transcricionais em tumores de pâncreas (Doctoral dissertation, Universidade de São Paulo).
- Pellegrina, D. V. D. S. (2016) Análise global da expressão de RNAs não codificadores no sistema imunológico humano na senescência e sepse (Masters dissertation, Universidade de São Paulo).

